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JavaScript For In

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The For In Loop

The for...in loop iterates over the enumerable properties of an object.

The for...in loop is primarily used for objects to access their property names (keys).

Syntax

```
for (key in object) {
  // code block to be executed
}
```

- **key**
  A variable that holds the name (key) of each property during the iterations
- **object**
  The object whose properties are being iterated over

A JavaScript `for in` statement loops through the properties of a person object:

Example

```
const person = {fname:"John", lname:"Doe", age:25};

let text = "";
for (let x in person) {
```

Example Explained

- The **for in** loop iterates over a **person** object
- Each iteration returns an **x** (a key)
- The x is used to access the **value** of x
- The value of x is **person[x]**

For In Over Arrays

The JavaScript `for in` statement can also loop over the properties of an Array:

Syntax

```
for (variable in array) {
  code
}
```

Example

```
const numbers = [45, 4, 9, 16, 25];

let txt = "";
for (let x in numbers) {
  txt += numbers[x];
}
```

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Control Flow

Like in other JavaScrpt loops, you can use control flow statements inside the loop:

Enumerable Properties

The `for...in` loop only iterates over enumerable properties.

Properties with the enumerable attribute is set to false, like some built-in methods or properties defined with `Object.defineProperty()` will not be included.

The Prototype Chain

The `for...in` loop also iterates over enumerable properties inherited from the object's prototype chain. To avoid this, you can use `hasOwnProperty()` inside the loop to check if the property belongs directly to the object itself.

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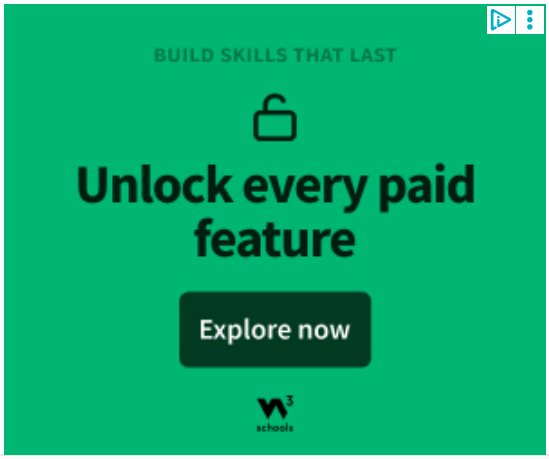

Note

Do not use **for in** over an Array if the **index order** is important.

The index order is engine-dependent, and array values may not be accessed in the order you expect.

It is better to use a **for** loop, a **for of** loop, or **Array.forEach()** when the order is important.

Array.forEach()

```
const numbers = [45, 4, 9, 16, 25];

let txt = "";
numbers.forEach(myFunction);

function myFunction(value, index, array) {
  txt += value;
}
```

Try it Yourself »

Note that the function takes 3 arguments:

- The item value
- The item index
- The array itself

The example above uses only the value parameter. It can be rewritten to:

Example

```
const numbers = [45, 4, 9, 16, 25];

let txt = "";
numbers.forEach(myFunction);

function myFunction(value) {
  txt += value;
}
```

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See Also:

The JavaScript for...of Loop

True or False.

The `for in` loop cannot be used on Arrays.

- True
- False

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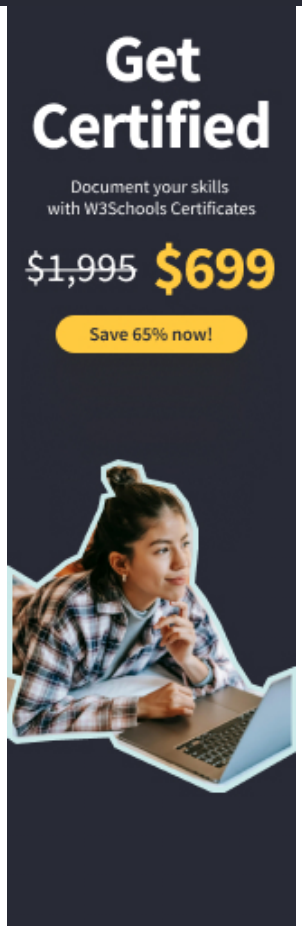

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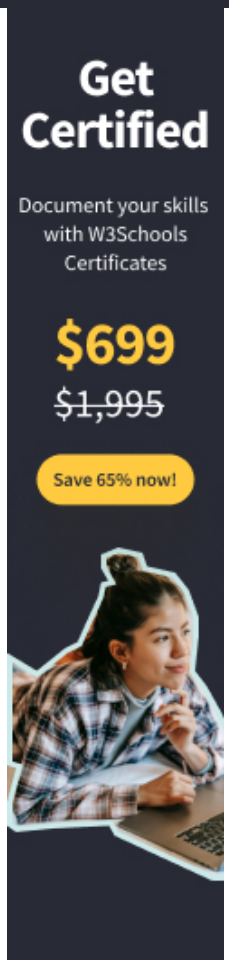

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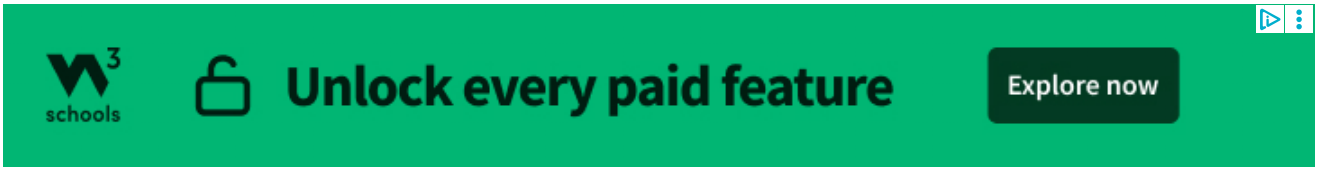

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